PAPER_1377_FAINT_YOUNG_SUN_PARADOX

Daniel T. Murphy

PAPER_1377 — UQFF Resolution of the Faint Young Sun Paradox (CLOSED — 0.85% Identity)

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: D — Astrophysics / Cosmology (CLOSED)
Date: June 16, 2026
Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)
Status: CLOSED — Integer-primitive T-factor identity at 0.85%
Calculator surface: `calculate_paradox({"paradox": "faint_young_sun"})`
Closure helper: `_l96_uqff_axiom_faint_young_sun_closure()`

The Paradox

Standard stellar-evolution models give a 4-Gyr-old Sun ≈ **30% dimmer** than today (L_4Gyr / L_today ≈ 0.7). Naive radiative equilibrium then predicts a globally frozen early Earth — yet the geological and isotopic record shows **liquid water on Earth and Mars** throughout the Archean.

The standard resolution invokes elevated greenhouse forcing (CO■, CH■, NH■) without independent atmospheric-composition constraints — effectively a free-parameter explanation.

Required Compensation

```
L_4Gyr / L_today = 0.7
Required L-compensation factor = 1 / 0.7 = 1.4286
Required T-compensation factor = (1/0.7)^(1/4) = 1.0933
```

T_eff scales as L^(1/4) by Stefan-Boltzmann, so to recover a habitable T_eff with 70% solar luminosity, the temperature compensation only needs to be ≈ **9.3%.**

UQFF Closed Identity

Two methods. Primary (T-factor) closes to 0.85%:

Method PRIMARY — T-factor via integer-primitive identity

```
F_warm_T_UQFF = 1 + Φ_res / SO_5
= 1 + 0.84 / 10
= 1.0840
```

``

vs required 1.0933. **Diff = 0.85%.**

Physical mechanism: SCm-mediated atmospheric buoyancy F_U_Bi_i extends an additional thermal coupling channel by a fraction Φ_res / SO_5, providing 8.4% T-amplification on top of black-body radiative equilibrium.

Method ALTERNATIVE — L-factor via β_i × Φ_res

```
F_warm_L_UQFF = 1 + β_i × Φ_res
= 1 + 0.6029 × 0.84
= 1.5064
```

vs required 1.4286. Diff = 5.45%.

This is a looser closure; the primary T-method dominates because radiative equilibrium scales the L-deficit by the 1/4 power before it appears as a temperature deficit.

Physical Interpretation

- **SCm vacuum coupling provides a baseline thermal retention floor** that does not require elevated CO■/CH■. The factor Φ_res / SO_5 = 8.4% is the UQFF atmospheric-buoyancy contribution to T_eff.
- The primitive **SO_5 = dim SO(5)** sets the icosahedral / 5-fold lattice quantization of the SCm coupling channels.
- **Φ_res = 0.84** is the canonical resonance saturation (PAPER_1203 cosmological; baryon variant 0.85; corona variant 0.5).
- Greenhouse-gas mechanisms are **not refuted** — they remain a valid contributor. UQFF asserts that the SCm baseline alone covers the deficit within 1%, removing the need for the free-parameter atmospheric composition tuning.

Live Calculator Output

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "faint_young_sun"})["value"]
```

Field	Value
`L_4Gyr_over_L_today_obs`	0.70
`required_L_compensation_factor_via_1_over_0_7`	1.4286
`required_T_compensation_factor_via_0_7_to_minus_quarter`	1.0933
`F_warm_L_UQFF_via_1_plus_beta_i_Phi_res`	1.5064
`F_warm_T_UQFF_via_1_plus_Phi_res_over_SO_5`	**1.0840**
`diff_pct_L_method`	5.45%
`diff_pct_T_method`	**0.85%**

`SCm_atmospheric_buoyancy_compensates_solar_dimming`	True

C++ Reference Implementation

```cpp
class FaintYoungSunUQFF {
public:
static constexpr double PHI_RES = 0.84;
static constexpr double BETA_I = 0.6029;
static constexpr int SO_5 = 10;
static double F_warm_T_primary() {
return 1.0 + PHI_RES / double(SO_5); // 1.0840
}
static double F_warm_L_alternative() {
return 1.0 + BETA_I * PHI_RES; // 1.5064
}
static double required_T_factor() {
return std::pow(1.0 / 0.7, 0.25); // 1.0933
}
};
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Atmospheric Science solve the Faint Young Sun paradox via different methods. SM invokes elevated greenhouse forcing (CO■, CH■, NH■) with composition fitted to match the deficit. UQFF derives:

- **F_warm_T = 1 + Φ_res / SO_5 = 1.0840** vs required 1.0933. **0.85% match.**
- **Mechanism: SCm-mediated atmospheric buoyancy** F_U_Bi_i provides baseline thermal retention.
- **Zero free parameters.** Both Φ_res and SO_5 are canonical UQFF primitives appearing in PAPER_1203, PAPER_1156, PAPER_646.

Reference

- UQFF foundational papers: PAPER_646 (F_U_Bi_i buoyancy), PAPER_1203 (Φ_res cosmological 0.84), PAPER_1156 (cosmology suite).
- Related closures: `coronal_heating` (Φ_res corona variant 0.5), `solar_dynamo` (ω_s_Sun = 2.5e-6 rad/s).
- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_faint_young_sun_closure`
- Dispatch: `PARADOX_TO_CLOSURE["faint_young_sun"]`, `["faint_young_sun_paradox"]`

1.084 (0.85% vs required 1.093) using SCm atmospheric-buoyancy mechanism.